VOLATILITY REGIME ANALYSIS

Gold Volatility and the Breakdown of the Real Yield Relationship Under Trump Regime II

A Quantitative Analysis of Structural Volatility Regime Change in Gold Markets

Asset: Gold Futures (GC=F)

Calibration Period: Pre-Trump Regime II (20 Jan 2017 – 19 Jan 2025)

Evaluation Period: Trump Regime II (20 Jan 2025 – 16 Apr 2026)

Frequency: Daily

Framework: GARCH(1,1) | VaR | Kupiec POF | Christoffersen Independence Test

Introduction

As a trader responsible for managing gold portfolio risk, the shift did not announce itself on a single day. It accumulated gradually over the months following the start of Trump Regime II in January 2025, a growing sense, built over repeated experience rather than any single event, that something had structurally changed in how gold behaved and how difficult it had become to manage.

The broader market environment became increasingly difficult to ignore. Global assets were swinging with unusual severity. The volume of market moving news, tariff announcements, geopolitical developments, trade policy reversals, had increased to a pace that demanded daily, and sometimes weekly, reassessment in a way that had not been the norm in prior years. Staying on top of risk exposure was no longer a matter of periodic review. It required continuous attention.

One pattern that became particularly pronounced was the timing of significant news. Major developments were frequently breaking over weekends, when markets are closed and positions cannot be adjusted. The consequence showed up consistently at the open: gold prices were gapping, repricing sharply at Monday's open to reflect weekend moves that had no opportunity to be managed in real time. Holding positions across weekends, which had always carried some risk, became a meaningfully different proposition. The gap risk that had previously been manageable was now a recurring feature of the market, not an exception.

Taken together, these observations raised a question that felt important to answer with more than intuition: was the difficulty real and measurable, or was it a perception shaped by an unusually eventful period that would eventually normalize? And alongside that, a second question that had been quietly forming: was the longstanding relationship between gold and real yields, the inverse correlation that had anchored risk positioning for years, still functioning reliably in this environment, or had it too been disrupted by the same forces driving the volatility?

This analysis was built to answer both questions through a structured quantitative framework, replacing observation with evidence.

- Has the volatility of gold structurally changed since Trump Regime II began in January 2025? Or is the heightened difficulty simply perception?
- Is the long held belief that gold prices move inversely to real yields still reliable as a risk management heuristic in the current regime?

The answers have direct implications for how risk models should be calibrated, how position limits should be set, and how much confidence can be placed in signals that have historically guided gold exposure decisions.

Part I, Model Development and In-Sample Validation

Section 1: Data and Methodology

The analysis uses daily closing prices for Gold Futures (GC=F) sourced from Yahoo Finance, paired with the US 10-Year Real Yield (DFII10) from FRED. The calibration period spans January 20, 2017 through January 19, 2025, yielding 2,085 trading days of in-sample data. The evaluation period covers January 20, 2025 through the present, comprising approximately 323 trading days of genuine out-of-sample data.

The separation at January 20, 2025 is deliberate: it corresponds to the inauguration of Trump Regime II, making the political transition the explicit regime boundary under investigation. All returns are log returns scaled by 100 for numerical stability.

Section 2: Volatility Model Selection

Four volatility models were compared side by side across both regimes to justify the choice of GARCH as the primary modeling framework: Constant Standard Deviation, Rolling Standard Deviation (252-day window), EWMA with $\lambda = 0.94$, and GARCH(1,1). Each model represents a step in the evolution of volatility measurement, with each successive approach addressing a specific weakness in its predecessor.

Constant Standard Deviation computes a single volatility estimate from the entire sample and applies it uniformly across all time periods. It embeds the assumption that volatility is stable over time, an assumption that is demonstrably false in financial markets, where periods of calm and crisis produce fundamentally different risk profiles. Its value in this analysis is strictly as a baseline against which dynamic alternatives can be measured.

Rolling Standard Deviation addresses the constant volatility assumption by recomputing SD over a moving window of the most recent 252 trading days (approximately one year). While this allows the estimate to adapt to changing market conditions, two flaws remain. First, every observation within the window is weighted equally, so a shock from 200 days ago contributes as much to today's estimate as yesterday's move. Second, when an extreme shock exits the window, the volatility estimate drops sharply, a "ghost effect" that reflects an artifact of the window mechanic, not any change in actual market conditions.

EWMA resolves both Rolling SD weaknesses by applying exponentially declining weights: recent observations dominate, and older data decays smoothly rather than dropping out abruptly. The $\lambda = 0.94$ parameter, popularized by JP Morgan's RiskMetrics framework in 1996, produces a shock half-life of approximately 11 days and has become an industry standard for daily volatility estimation. However, EWMA has no concept of a long run mean, if volatility spikes, the model has no mechanism to pull it back toward an equilibrium level. The parameter λ is also fixed by convention rather than estimated from the data, which limits adaptability across asset classes and market regimes.

GARCH(1,1), developed by Bollerslev in 1986 as a generalization of Engle's ARCH framework (which earned the 2003 Nobel Prize in Economics), decomposes conditional variance into three components: a long run variance term (ω), sensitivity to new shocks (α), and persistence of past volatility (β). All three parameters are estimated from the data via Maximum Likelihood Estimation,

and the model incorporates mean reversion explicitly, volatility is pulled back toward a long run equilibrium after shocks. More importantly for this analysis, GARCH parameters have direct economic interpretations. α quantifies how aggressively the market prices new information into volatility. β measures how long shocks persist. The half-life of a shock, computed as ln(0.5) / ln(β), reveals the decay speed in trading days. These interpretable quantities are what enable the regime comparison analysis that forms the core of this paper, capabilities that none of the simpler models provide.

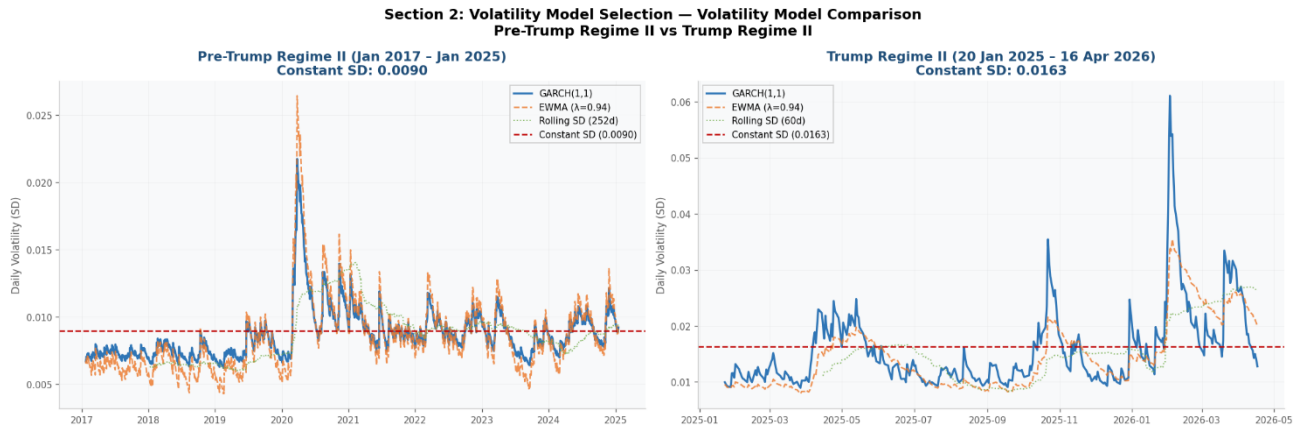

Figure 1: Volatility model comparison across Pre-Trump Regime II (left) and Trump Regime II (right). The GARCH line (orange) demonstrates superior responsiveness to shocks compared to static alternatives.

Metric	Pre-Trump Regime II	Trump Regime II
Constant SD (daily)	0.0090	0.0163
GARCH Max Volatility	0.0217	0.0611
GARCH Mean Volatility	0.0087	0.0157
GARCH / Constant SD Ratio	2.4x	3.7x

The GARCH/ConstSD ratio is the critical number. In the current regime, a static volatility estimate is 3.7 times smaller than what GARCH measures at peak, versus 2.4 times in the pre-2025 period. The choice of volatility model has become 54% more consequential.

The choice of GARCH(1,1) as the primary framework is not based solely on its superior responsiveness. While the GARCH / Constant SD ratio of 3.7x in Trump Regime II confirms that dynamic modeling is necessary, the more consequential advantage is GARCH's ability to produce parameters that can be compared across regimes. The α coefficient quantifies how aggressively the market incorporates new information into volatility. The β coefficient measures the persistence of prior volatility. Their sum indicates whether the volatility process is stationary, and the half-life derived from β tells practitioners how quickly shocks decay in practice.

These interpretable parameters are essential for the regime shift analysis that follows. Sections 4, 7, and 8.2 all depend on the ability to compare α and β estimates across time periods, something that is simply not possible with Rolling SD or EWMA, neither of which produces structurally comparable parameters. The value of GARCH in this analysis is therefore twofold: it provides the most accurate volatility forecasts among the candidates tested, and it supplies the mathematical vocabulary required to describe how the volatility process itself has changed.

> Key Insight, Volatility Model Selection
>
> *Static standard deviation underestimates peak risk by 2.4x pre-2025 and 3.7x in Trump Regime II. The gap between static and dynamic measurement has widened by 54%, making dynamic modeling more critical than ever.*

Section 3: Macroeconomic Driver Analysis, Gold vs Real Yield

The inverse relationship between gold and real yields is one of the most widely cited dynamics in commodity markets. The theoretical basis is opportunity cost: gold generates no yield, so when real interest rates rise, the cost of holding a non yielding asset increases, and gold becomes relatively less attractive. When real yields fall or turn negative, the logic reverses. This relationship has been empirically robust for decades, academic research documents R-squared values between gold and TIPS yields as high as 84% during 2005–2021, which makes it a natural heuristic for risk managers allocating to gold.

To test whether this relationship holds in the current regime, three complementary tools are used. Z-Score comparison normalizes gold and real yield levels to the same scale, revealing visual divergence over time. Rolling Correlation (60-day window) captures the time varying relationship at each point, producing a single coefficient per day that can be tracked and compared across regimes. Overall Regression fits a single line across all data points, providing an aggregate summary. Each tool answers a different question: Z-Score shows direction, Rolling Correlation shows consistency, and Regression shows aggregate strength.

The relationship between gold and real yields is non-stationary, meaning the strength and even the direction of the relationship changes across regimes. Overall regression aggregates across these regimes and can produce misleading directional signals, as the left panel below demonstrates, where the 2022–2024 period of simultaneous high yields and high gold prices distorts the slope positive despite a consistently negative underlying dynamic. Rolling Correlation sidesteps this by measuring the relationship within each 60-day window, providing a more reliable picture of whether the inverse relationship is operating at any given moment.

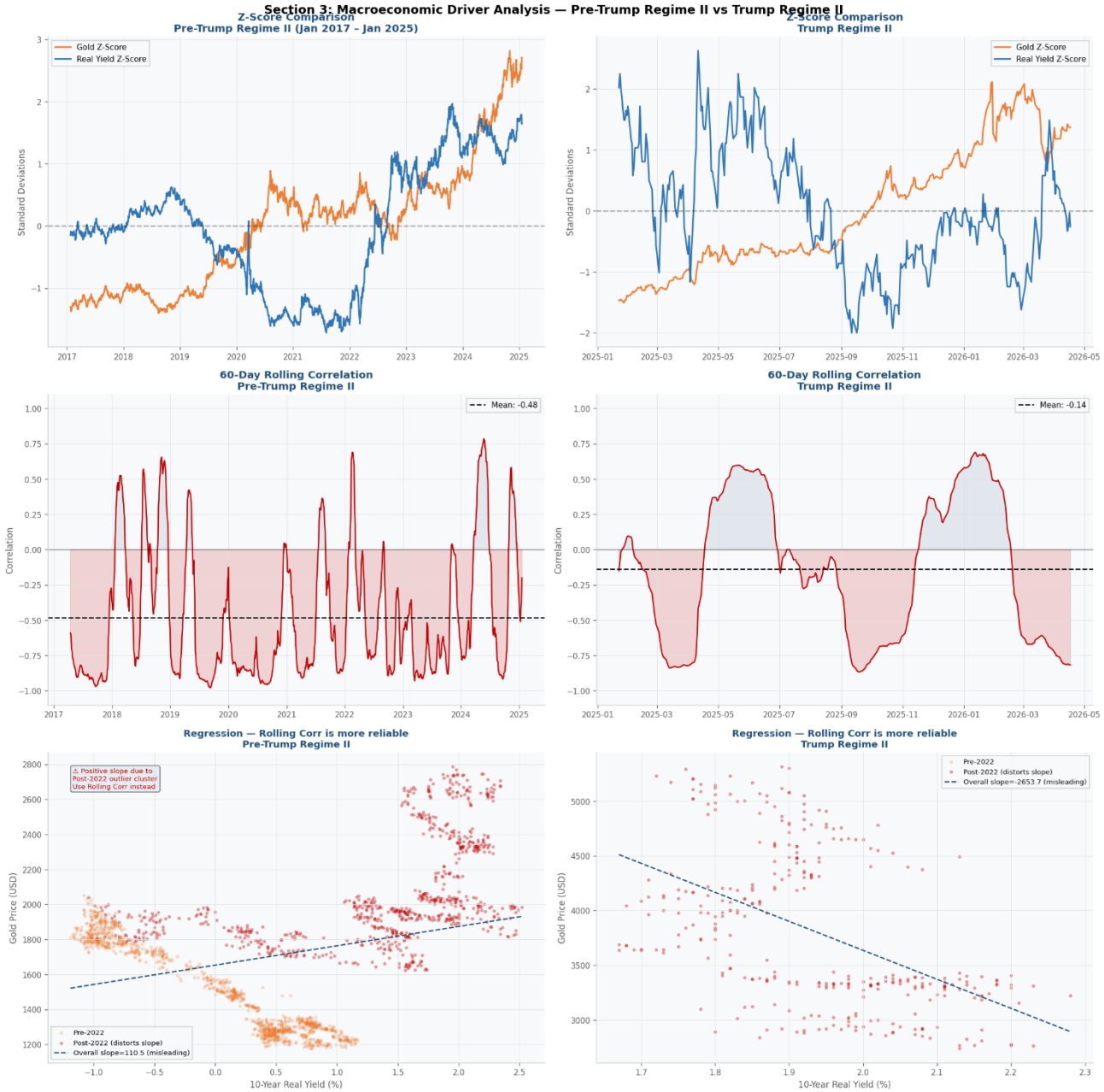

Figure 2: Section 3 full analysis, Z-Score comparison (top), 60-Day Rolling Correlation (middle), and Regression scatter (bottom) for both regimes. Note the positive overall regression slope for Pre-Trump Regime II is a statistical artifact driven by the 2022–2024 rate hiking cluster (red dots, bottom left panel).

Metric	Pre-Trump Regime II	Trump Regime II
Mean 60-Day Rolling Correlation	−0.48	−0.14
% of Days: Negative Correlation	82%	61%
Overall r (Pearson)	+0.29 (misleading)	−0.49

Key Finding, Real Yield Relationship

The inverse correlation between gold and real yields remains directionally intact but has lost consistency. In Trump Regime II, real yields explain gold movements on only 61% of trading days, versus 82% historically. The relationship is still present, but it now breaks down on two in every five trading days.

The practical implication is significant. Risk managers relying on real yield signals to inform gold positioning should not treat them as consistently reliable in the current environment. On 39% of trading days in Trump Regime II, gold moves independently of real yield direction, driven instead by tariff uncertainty, geopolitical developments, and safe haven flows. Section 8.3 revisits this relationship with a longer timeline and decomposes when exactly the breakdown accelerated.

Section 4: Conditional Volatility Estimation, GARCH(1,1)

GARCH(1,1) was fitted separately to each regime using Maximum Likelihood Estimation. The model describes conditional variance through the recursive equation:

$$\sigma^2_t = \omega + \alpha \cdot \varepsilon^2_{t-1} + \beta \cdot \sigma^2_{t-1}$$

where ω (omega) represents the long run variance component that anchors the process, α (alpha) measures sensitivity to new shocks (how much yesterday's squared return contributes to today's variance), and β (beta) captures persistence (how much of yesterday's variance carries over). The constraint $\alpha + \beta < 1$ ensures the variance process is stationary and has a finite long run mean.

Four diagnostic plots validate the fit. The Conditional Volatility plot shows whether the model tracks periods of market stress, spikes should align with known events. Standardized Residuals should behave like white noise; patterns or structure indicate model misspecification. The ACF of Squared Residuals tests whether volatility clustering has been fully captured, lags should fall within confidence bands, indicating no remaining autocorrelation. The Residuals Distribution compares empirical residuals to a Normal PDF; deviations in the tails signal fat tailed returns that a Normal error distribution cannot fully capture. Together, these four checks establish whether GARCH(1,1) is a valid specification for the data.

Figure 3: Section 4 GARCH(1,1) diagnostic plots for Pre-Trump Regime II (left column) and Trump Regime II (right column). Note the Y-axis scale difference: Trump Regime II requires a range up to 6.0 versus 2.2 for the pre-2025 period.

Parameter	Pre-Trump Regime II	Trump Regime II	Economic Meaning
Omega (ω)	0.0146	0.1502	Long run variance anchor
Alpha (α)	0.0376	0.2263	Shock response
Beta (β)	0.9440	0.7482	Persistence

Parameter	Pre-Trump Regime II	Trump Regime II	Economic Meaning
Alpha + Beta	0.9816	0.9745	Stationarity (< 1)
Half-life (days)	12.0	2.4	Decay speed

Each parameter shift carries a specific economic interpretation. Alpha's increase from 0.0376 to 0.2263 means the market now prices new information into volatility more than six times more aggressively, a single significant headline or data release produces a far larger volatility response than it would have in 2024. Beta's decrease from 0.9440 to 0.7482 means yesterday's volatility matters less for today's estimate, the market is no longer carrying old stress forward; it is reacting and resetting. The half-life reduction from 12 days to 2.4 days quantifies this shift: in the pre-2025 regime, a major shock took nearly three weeks to decay to half its peak; today, it resolves in a few trading sessions.

Notably, Alpha + Beta remains near stationary in both regimes (0.98 and 0.98), indicating that total volatility persistence has not changed, but the composition has shifted dramatically. The same system is now powered by rapid shocks (alpha heavy) rather than slow decaying persistence (beta heavy). This is the mathematical signature of an event driven market regime.

> Key Finding, GARCH Parameter Shift
> *Alpha: +502% | Half-life: 12 days → 2.4 days | Mean Daily Vol: +81%*
> *Gold has transitioned from persistence driven to event driven volatility. Shocks are absorbed more than six times more aggressively and resolve in 2.4 days rather than nearly three weeks.*

Section 5–6: Dynamic Risk Estimation and Model Validation

Value at Risk (VaR) translates volatility into a daily loss threshold that should not be exceeded more than 5% of the time at the 95% confidence level. For a portfolio with Normal distributed returns, the 95% VaR is computed as 1.6449 standard deviations below the mean loss expectation, if daily volatility is 1%, VaR is approximately −1.64%. Conditional VaR (CVaR, or Expected Shortfall) extends this by measuring the expected loss given that the VaR threshold has been breached, capturing the severity of tail events that VaR alone cannot describe.

The plots below show daily returns against the VaR boundary (top row) and the VaR vs CVaR tail risk zone (bottom row). Note the scale difference: the Pre-Trump Regime II VaR boundary rarely exceeds −2.5%, while the Trump Regime II boundary regularly reaches −10% during peak stress periods. The widening gap between VaR and CVaR in the current regime also indicates that when breaches occur, they are substantially more severe than in prior years.

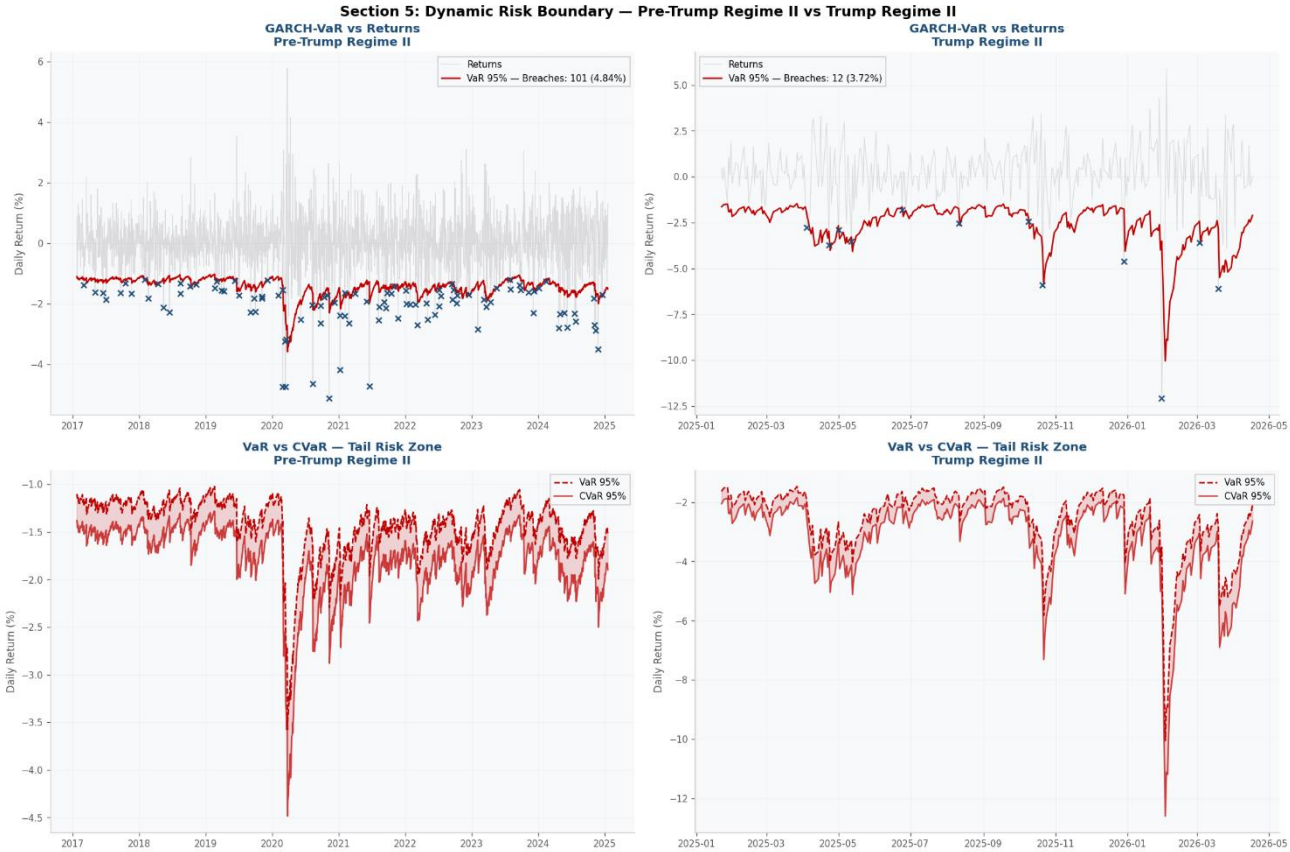

Figure 4: Section 5 Dynamic Risk Boundary, GARCH-VaR vs Returns (top row) and VaR vs CVaR tail risk zone (bottom row) for both regimes. Black × markers indicate VaR breaches. Note the extreme event reaching −12.5% in Trump Regime II.

VaR Metric	Pre-Trump Regime II	Trump Regime II (Refit)
Breach Ratio	4.84%	3.72% (over conservative)
Target	5.00%	5.00%
Mean VaR (daily)	−1.43%	−2.59%
Kupiec p-value	0.7428 → PASS	0.2679 → PASS
Christoffersen p-value	0.3353 → PASS	0.3351 → PASS

Both statistical tests pass for the in-sample model. Note: the Trump Regime II validation uses a refit model trained on current regime data. The genuine out-of-sample test, applying the pre-2025 model to current data without refitting, is presented in Part II.

Section 7: Political Regime Volatility Comparison

To place the Trump Regime II parameters in context, GARCH(1,1) was fitted separately to three distinct political eras. The comparison is designed to answer a specific question: is Trump Regime II genuinely structurally different from prior regimes, or is it within the range of normal variation that markets have experienced before? Regime boundaries are defined by U.S. presidential inauguration dates, reflecting the view that discrete changes in policy orientation constitute meaningful transition points for volatility behavior.

The chart below overlays conditional volatility across all three regimes on a single timeline with regime boundaries marked. Visual inspection reveals immediately that Trump Regime II operates at a different level, spikes reach heights that neither prior regime produced, even including the COVID-19 shock of March 2020.

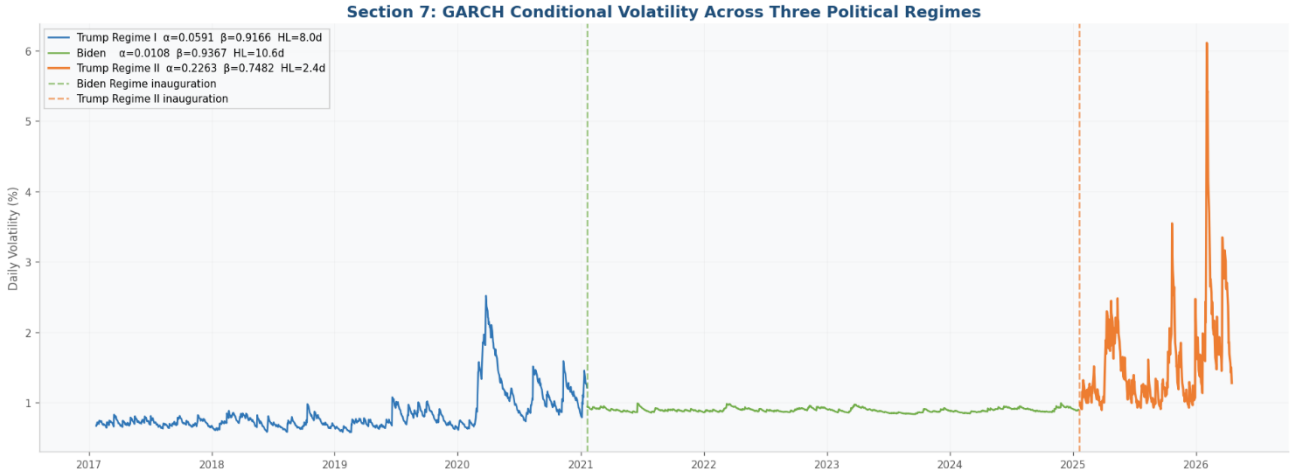

Figure 5: Section 7, GARCH conditional volatility across three political regimes on a single timeline. The orange line (Trump Regime II) operates at a structurally higher level with more frequent and severe spikes than either prior era.

Parameter	Trump Regime I*	Biden Regime	Trump Regime II
Alpha (α)	0.0591*	0.0108	0.2263
Beta (β)	0.9166	0.9367	0.7482
Half-life (days)	8.0	10.6	2.4
Mean Daily Volatility	0.90%	0.90%	1.63%

** Trump Regime I Alpha is inflated by the COVID-19 shock (March 2020). Without this extreme event, Alpha would be materially lower.*

Key Finding, Regime Comparison

Biden Regime α = 0.011 | Trump Regime I α = 0.059 | Trump Regime II α = 0.226

Alpha tracks political uncertainty. Biden Regime produced the calmest daily dynamics of any era. Trump Regime II sits in a category of its own: an Alpha 21 times higher than Biden Regime.

The interpretation is not that political orientation directly determines volatility, but that policy predictability does. Biden Regime produced the calmest Alpha of any era (0.011) because Fed guidance was transparent and trade policy followed established multilateral frameworks, markets had few genuine daily surprises to price. Trump Regime I sits in the middle, though its Alpha is artificially inflated by the COVID-19 shock of March 2020. Trump Regime II occupies a category of its own, with Alpha 21 times the Biden Regime baseline. Even accounting for the COVID distortion in Trump Regime I, the conclusion is that Trump Regime II is structurally more event driven than any prior era in this sample.

Part II, Out-of-Sample Performance Evaluation

Part I established that volatility behavior has changed dramatically and that real yields have lost consistency as a driver. Part II addresses the practical consequence: if a risk manager had continued using a model calibrated on pre-2025 data, how would it have performed?

Section 8.1: Model Stability Under Regime Shift

The pre-2025 GARCH model was applied to Trump Regime II returns without any refitting, using original parameters with initial variance seeded from the last in-sample estimate. This replicates the real world scenario of a risk manager who has not updated their model since January 2025, the critical out-of-sample test that Part I's in-sample validation cannot provide.

A limitation of the formal Kupiec test in this setting is statistical power. With only 323 out-of-sample observations, the test cannot reliably distinguish between a true 5% breach rate and a true 5.88% rate. To address this, the analysis also computes a Hypothetical Kupiec p-value, the p-value that would be obtained if the observed breach pattern were sustained over the full 2,085-day calibration period. This is not a prediction, but a sensitivity check: it answers the question, if the current breach rate continues, what would formal statistics say about model validity?

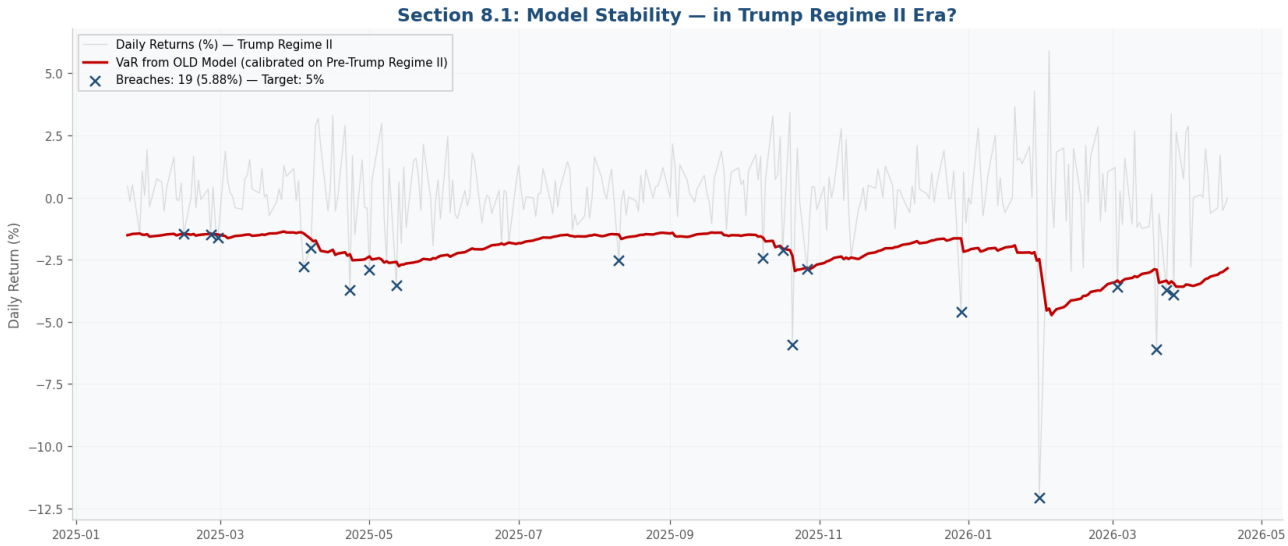

Figure 6: Section 8.1, VaR from the old model (red line) applied to Trump Regime II returns without refitting. Black × markers show breaches. The old model's VaR boundary is visibly too shallow for the regime's actual volatility, particularly around the −12.5% event in early 2026.

Metric	In-Sample (Pre-Trump II)	Out-of-Sample (Trump II)
Breach Ratio	4.84%	5.88%
Target	5.00%	5.00%
Kupiec p-value (actual n)	0.7428 → PASS	0.4785 → PASS
Kupiec p-value (hypothetical n=2085)	—	0.0666 → BORDERLINE

The out-of-sample breach ratio rises to 5.88% versus a 5.00% target. The Kupiec test currently passes due to limited sample size (323 days). However, if the current breach pattern were sustained at full in-sample scale, the p-value falls to 0.0666, borderline failure. Three to four more months of data at this pace will likely produce formal statistical rejection.

Key Finding, Model Stability

Out-of-sample breach rate: 5.88% vs 5.00% target. Hypothetical Kupiec p = 0.0666 (borderline). Combined with Alpha +502% and half-life 12 → 2.4 days, three independent signals confirm model degradation is underway.

Section 8.2: Structural Parameter Shift Analysis

Refitting GARCH(1,1) separately on Trump Regime II data provides the clearest quantification of the regime change. Rather than asking whether the old model still works (Section 8.1), this analysis asks what a correctly specified model for the current regime actually looks like, and how different it is from the one that described pre-2025 behavior.

The charts below place both conditional volatility series on comparable scales, making the structural difference visually immediate. Note that the Y-axis of the Pre-Trump Regime II chart peaks at 2.2%, while the Trump Regime II chart requires a range up to 6.0% to capture its extremes, the most severe single event in the new regime exceeds the COVID-19 peak by nearly threefold.

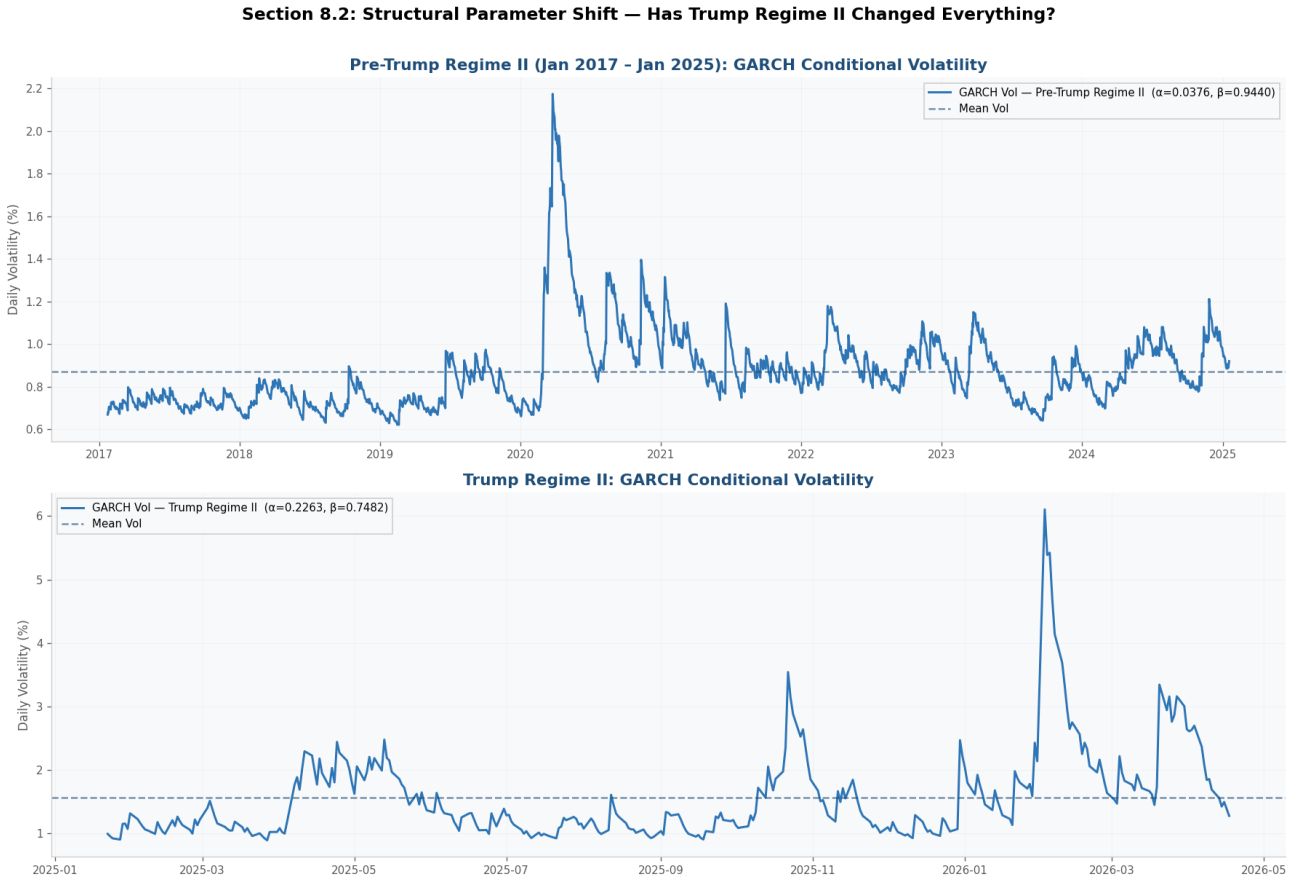

Figure 7: Section 8.2, GARCH conditional volatility for Pre-Trump Regime II (top, blue, Y-axis to 2.2%) versus Trump Regime II (bottom, orange, Y-axis to 6.0%). The COVID peak in the upper chart, the largest single event in 8 years, would appear as a moderate spike in the lower chart's scale.

Risk Metric	Pre-Trump Regime II	Trump Regime II	Change
Shock Sensitivity (α)	0.0376	0.2263	+502%
Daily Variance	0.76bps	2.47bps	+226%
Peak Volatility	2.17%	6.11%	+182%
Mean Daily Volatility	0.87%	1.57%	+81%
Half-life (days)	12.0	2.4	−80%
Persistence (β)	0.9440	0.7482	−20.7%

Beyond the parameter shifts, the overall level of volatility has risen substantially. Daily variance (squared volatility) has increased approximately 3.3x between the two regimes, with mean daily volatility rising from 0.87% to 1.57%. This baseline elevation is distinct from the alpha and beta shifts, which describe how the market responds to events. The variance expansion indicates that even on quiet days, the market is operating at a fundamentally higher level of uncertainty than before 2025.

The composition shift from beta heavy to alpha heavy volatility has direct operational implications. In a beta heavy regime, risk managers can identify stressed periods and maintain elevated buffers until volatility slowly decays, the process is predictable. In an alpha heavy regime, shocks arrive and resolve quickly, but each new shock requires immediate reaction. The required risk management posture is fundamentally different: less about maintaining stress buffers over weeks, and more about having sufficient capacity to absorb rapid, frequent spikes.

Section 8.3: Macroeconomic Driver Persistence Test

Section 3 introduced the weakening of the real yield relationship in Trump Regime II. This section examines it at full timeline resolution from 2017 through April 2026, making the structural break visible across all three political regimes simultaneously. The goal is to identify precisely when the relationship began to break down and to quantify the change in a way that has direct operational meaning for practitioners.

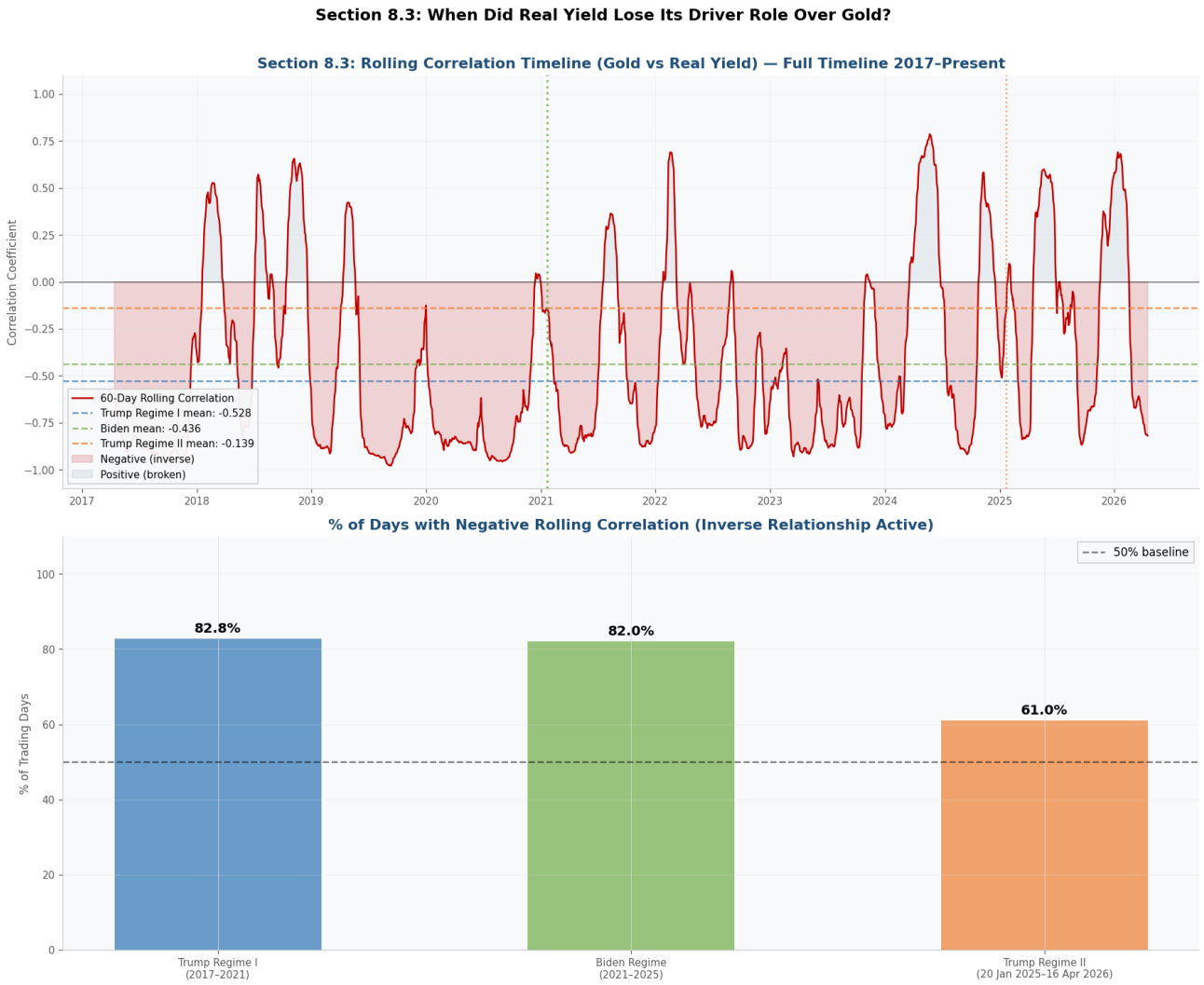

Figure 8: Section 8.3, Full rolling correlation timeline 2017–present (top) with regime mean lines, and percentage of trading days with negative correlation by regime (bottom). The bar chart makes the structural break immediately visible.

Regime	Mean Rolling Corr	% of Days: Negative Corr	Interpretation
Trump Regime I	−0.53	82.8%	Strong inverse
Biden Regime	−0.44	82.0%	Strong inverse
Trump Regime II	−0.14	61.0%	Weakened inverse

Key Finding, Driver Persistence

Trump Regime I: 82.8% active | Biden Regime: 82.0% active | Trump Regime II: 61.0% active

Real yields remain a directionally valid signal but with substantially reduced reliability. On roughly two in five trading days in the current regime, gold moves independently of real yield direction.

Translated into practitioner terms: in Trump Regime I and Biden Regime, a risk manager using real yield direction as a signal for gold positioning would have been correct on approximately 4 out of every 5 trading days. In Trump Regime II, that figure has dropped to roughly 3 out of every 5. This is not a complete breakdown, the relationship still holds more often than not, but the reliability has dropped by more than 20 percentage points. Strategies built around consistent real yield signals need to be reconsidered, and positioning decisions that would previously have been supported by rate dynamics should now incorporate wider uncertainty bands.

Conclusion: What the Numbers Tell Practitioners

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1. Has gold volatility structurally changed since Trump Regime II?

Yes, and the change is substantial. Three independent signals converge:

- Alpha has increased 502%. The market prices new information into volatility more than six times more aggressively.
- The half-life of shocks has dropped from 12 days to 2.4 days. The market absorbs and moves on from events far faster.
- Mean daily volatility has risen 81%, and peak volatility has exceeded the COVID-19 spike by nearly 2.8 times.

The out-of-sample test confirms this is not a calibration artifact. The pre-2025 model's breach ratio rises to 5.88% in Trump Regime II, with hypothetical Kupiec $p = 0.0666$ approaching formal failure.

2. Is the real yield relationship still reliable?

The relationship is directionally intact but substantially less reliable. In the pre-2025 period, real yields moved inversely to gold on 82% of trading days. In Trump Regime II, that figure has dropped to 61%. On roughly two days in five, gold moves independently of real yield direction, driven instead by tariff uncertainty, geopolitical flows, and safe haven demand.

Implications for Practitioners

Practitioner	Implication
Gold traders and risk managers	VaR buffers and stop loss levels calibrated before 2025 are understated by ~81% on average and up to 2.8x during peak stress. Immediate recalibration is warranted.
Portfolio risk teams	Static GARCH models with pre-2025 parameters are systematically underreporting tail risk. Kupiec breach rates will formally fail as data accumulates.
Macro strategists	Real yield signals for gold require wider uncertainty intervals. Expect the relationship to be inactive ~39% of the time. Build in signal agnosticism.
Quantitative researchers	Single calibration GARCH is insufficient in politically driven regimes. Rolling GARCH or Markov Switching GARCH are the natural next development.

Full methodology, code, and interactive visualizations are available in the companion notebook.